

Fictitious play with policy gradient on soccer

Longer runs, more seeds, and whether the early policies slow the average

Summary

- PPO fictitious play (random starting states, state-average opponent) on the 5x3 soccer board, run to 800 rounds on seeds 8-15. The exploitability of the average policy falls from 0.31 at round 100 to 0.17 at round 800 (mean over 8 seeds). It has not reached 0.
- 7 of the 8 seeds end at 0.22 or lower; seed 15 stays at 0.71. Without seed 15 the mean is 0.09.
- Testing the guess that the average declines slowly because it still contains the weak early policies: the guess is partly right. Dropping more of the early history lowers the mean exploitability, but the first quarter alone explains only a small part (0.17 to 0.13 at round 800), and for some seeds dropping early policies makes the average worse.

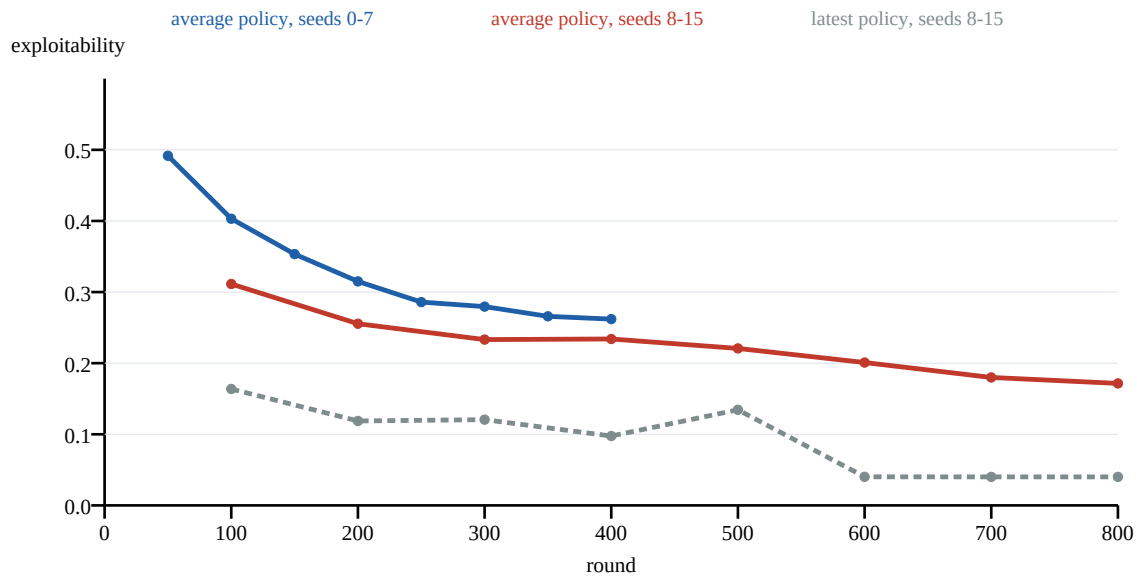
Setup

Soccer is a deterministic two-player zero-sum grid game, solved exactly by minimax value iteration (discount 0.9, 100-step limit counted as a tie). Exploitability is the exact best-response gain against a policy, so 0 is an equilibrium. PPO learns a best response to the state-by-state average of the opponent's earlier policies, from random starting states, 20 iterations of 256 games per round. The average policy is the plain average of every snapshot so far. Seeds 0-7 were run to 400 rounds earlier; seeds 8-15 were run to 800 rounds here.

Longer run

Round	Average policy	Latest policy	Seeds with latest under 0.05
100	0.31 (0.09-0.55)	0.16 (0.00-0.46)	5 of 8
200	0.26 (0.04-0.48)	0.12 (0.00-0.49)	6 of 8
300	0.23 (0.03-0.49)	0.12 (0.00-0.49)	6 of 8
400	0.23 (0.02-0.72)	0.10 (0.00-0.78)	7 of 8
500	0.22 (0.02-0.73)	0.13 (0.00-0.62)	6 of 8
600	0.20 (0.02-0.73)	0.04 (0.00-0.32)	7 of 8
700	0.18 (0.02-0.73)	0.04 (0.00-0.32)	7 of 8
800	0.17 (0.02-0.71)	0.04 (0.00-0.32)	7 of 8

Exploitability, mean over seeds 8-15 with the range in brackets.

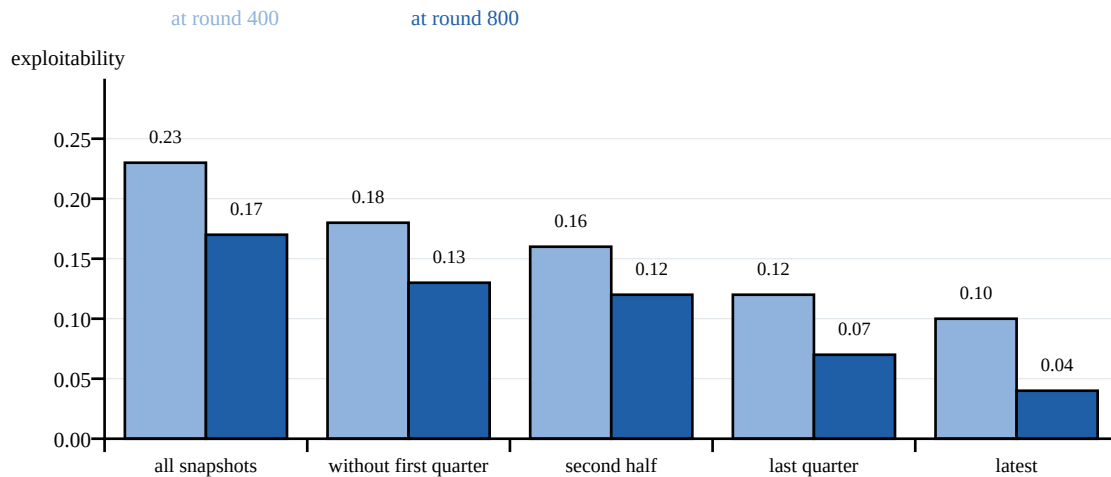


Mean exploitability by round. The average policy keeps falling after round 400 (0.23 to 0.17), about 0.015 per 100 rounds.

- Over all 16 seeds the average policy at round 400 is 0.25 on average (0.02-0.72), and 5 of the 16 are under 0.05.
- The latest policy is below 0.05 for 7 of 8 seeds at round 800, but earlier seeds showed it can jump (seed 0 went from 0.00 to 0.60), so fictitious play's claim is about the average policy.

Do the early policies slow the average?

Each checkpoint stores every snapshot so far, so the same snapshots can be averaged in different ways: all of them, without the first quarter, the second half only, the last quarter only, or just the latest. If the early policies were the main drag, the averages that drop them would be much less exploitable.



Mean exploitability over the 8 seeds under each way of averaging.

Way of averaging	At round 400	At round 800
all snapshots (the average policy)	0.23	0.17
without the first quarter	0.18	0.13
second half only	0.16	0.12
last quarter only	0.12	0.07
latest snapshot only	0.10	0.04

At round 800, per seed:

Seed	All	Without first quarter	Second half	Last quarter	Latest
8	0.13	0.04	0.00	0.00	0.00
9	0.22	0.10	0.11	0.00	0.00
10	0.02	0.01	0.01	0.02	0.00
11	0.07	0.03	0.00	0.00	0.00
12	0.05	0.00	0.00	0.00	0.00
13	0.11	0.14	0.18	0.00	0.00
14	0.06	0.08	0.13	0.26	0.00
15	0.71	0.67	0.54	0.32	0.32

- The mean falls steadily as more of the early history is dropped, at both rounds, which matches the guess in direction.
- The first quarter alone accounts for a small part: 0.23 to 0.18 at round 400 and 0.17 to 0.13 at round 800. At round 800 dropping it helps 6 of the 8 seeds and hurts 2 (seeds 13 and 14).
- It does not explain every seed. In seeds 13 and 14 the average gets worse when the early policies are dropped (seed 14: 0.06 for all snapshots, 0.26 for the last quarter). Seed 15 is poor under every way of averaging, so something other than the early policies is wrong there.

What this does and does not show

- It shows that the early policies add to the exploitability of the average policy, in most seeds.
- It does not test the explanation that they fade like 1 over the number of rounds, and it does not say why seed 15 is stuck.
- It is one checkpoint per seed, 8 seeds, one small board and one set of settings. The learning rate and entropy weight were tuned on seed 0.

Raw numbers are in results/soccer_fp_diagnosis_runs.jsonl, results/soccer_averaging_round_400.txt and results/soccer_averaging_round_800.txt. The averaging test is doggame/soccer_averaging.py.